

Prescribed Banzhaf Power by Exact Computation: Closed-Form Realizations and Certified Weighted Obstructions

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Abstract

Designing a voting rule that realizes a prescribed distribution of voting power — the inverse power index problem — is a fundamental computational task for parliaments, councils, and other weighted voting bodies. This paper develops an exact, fully reproducible computational methodology for the inverse Banzhaf problem and applies it to the egalitarian benchmark target of Kurz (2015), in which a single player holds half the voting power of each of the remaining players.

On the constructive side, a single parametric family of simple games (the W-family) with an exact closed-form swing formula yields explicit realizing games for every number of voters from six upward, resolving Kurz's Conjecture 16 and correcting the sole erroneous published example; a compact formulation with a cubic number of variables solves every instance up to eighty voters in seconds to minutes. On the obstruction side, a sharpened swap/block rigidity condition drives a deterministic constraint-programming satisfiability (CP-SAT) certification that no weighted majority game on six to ten voters realizes the target, at a conclusive weight bound, and the certified optimal weighted approximation is computed exactly for up to eight voters.

As a computational case study, the methodology is applied to the historical weighted Council of the European Union: its exact Banzhaf power distributions

are reproduced and shown to be far from the target, with the certified optimal redesign for six voters improving the distance about threefold. Every numerical claim is reproducible from the companion code package.

Keywords: voting games; Banzhaf power index; inverse power index problem; integer programming; constraint programming

1. Introduction: the inverse Banzhaf problem

Designing a voting rule for a committee of *dissimilar* members — a weighted parliament, the Council of the European Union, or a federal upper chamber — is a decision problem: the designer must decide how voting *power* is to be distributed among members. The Banzhaf index (and the related Shapley–Shubik index) quantify power as the normalized number of *swings*, i.e., coalitions in which a member is decisive (Penrose 1946; Felsenthal & Machover 1998); for a weighted majority game, computing these indices is NP-hard (Matsui & Matsui 2001). The **inverse power index problem** is the formal dual of this computation: given a desired power distribution, decide whether a voting rule realizes it, construct one if so, and certify impossibility otherwise (Nurmi 1982; Kurz & Napel 2014; de Keijzer et al. 2014). From an operations-research standpoint this is an *inverse-optimization* problem: the power index is the forward map from rules to power distributions, and the designer must invert it subject to the structural constraints of a chosen rule class — general simple games, or the weighted majority games used by most real voting bodies (Muroga 1971).

This paper contributes an exact, fully reproducible computational methodology for both halves of this design decision problem, and applies it to a benchmark target that has resisted a unified treatment since 2014: the vector $\psi^n = (2, \dots, 2, 1) / (2n - 1), (1)$ which assigns to $n - 1$ members twice the power of the remaining member. Kurz (2015) conjectured the existence of a realizing simple game for every $n \geq 6$:

Conjecture 16 (Kurz 2015). For every $n \geq 6$ there exists a simple game v on n voters such that $Bz(v) = \psi^n$, where Bz is the (normalized) Penrose–Banzhaf index.

and the natural dual — that no *weighted majority* game can do so (Kurz’s Conjecture 15). Kurz and Napel (2014) verified Conjecture 16 computationally for $6 \leq n \leq 18$, but gave no construction valid for all n ; we provide one (Corollaries

14–17, Section 11), and settle the weighted obstruction computationally for $n \leq 10$ (Section 9).

The paper is organized as an exact computational study, centered on two engines. On the constructive side, Sections 3–7 give closed-form families, exact swing formulas, and explicit games realizing ψ^n for every $n \geq 6$; Section 8 summarizes the constructive half. On the obstruction side, Section 9 derives a sharp structural rigidity property (M') and certifies by the constraint-programming satisfiability (CP-SAT) solver that no weighted majority game realizes ψ^n for $n \leq 10$; Section 10 collects the two engines as a reusable methodology, with complexity statements and instance tables; Section 11 develops the unified W-family framework, the exact closed-form constructions, and the open problems. Section 12 applies the methodology as an exact computational case study to the historical weighted Council of the European Union, whose actual Banzhaf power distributions fall exactly in the certified range. Section 13 concludes.

2. Preliminaries

Let $N = \{1, \dots, n\}$ be a set of voters. A **simple game** (Taylor & Zwicker 1999) is a monotone Boolean function $v : 2^N \rightarrow \{0, 1\}$ with $v(\emptyset) = 0$ and $v(N) = 1$. A coalition $S \subseteq N$ is **winning** if $v(S) = 1$ and **losing** otherwise. A winning coalition is **minimal winning** if all its proper subsets are losing; a simple game is uniquely determined by its family of minimal winning coalitions.

For $i \in N$, let $\beta_i(v) = |\{S \subseteq N \setminus \{i\} : v(S) = 0, v(S \cup \{i\}) = 1\}|$ be the number of **swings** of player i (the Banzhaf measure). The (normalized) **Penrose–Banzhaf index** is $Bz_i(v) = \beta_i(v) / \sum_j \beta_j(v)$. Thus $Bz(v) = \psi^n$ iff $\beta(v)$ is proportional to $(2, \dots, 2, 1)$, i.e., iff $n - 1$ players have an equal number $2c$ of swings and one player has c swings, for some integer $c \geq 1$.

3. The complement-of-edges family

For a graph G on vertex set N with edge set E , consider the simple game v_G whose minimal winning coalitions are exactly the complements of the edges of G : $v_G(S) = 1 \iff S \supseteq N \setminus \{a, b\}$ for some edge $\{a, b\} \in E$. Equivalently, a coalition is winning iff it misses no edge of G . We prove:

Theorem 1. Let G be a graph on n vertices with edge set E . Suppose (i) G has no isolated vertices, and (ii) G is not a star (no vertex is incident with every

edge). Then the Banzhaf swing counts of v_G are $\beta_i(v_G) = |E| + (n-1) - 2 \cdot \deg_G(i)$ ($i \in N$). (2)

Proof. Player i has a swing on a losing coalition S with $i \notin S$ and $S \cup \{i\}$ winning. The latter means $S \cup \{i\} \supseteq N \setminus \{a, b\}$ for some edge $\{a, b\}$. - If $|S| = n-3$, then $S \cup \{i\} = N \setminus \{a, b\}$ is an $(n-2)$ -set, so $S = N(\{a, b\} \cup \{i\})$ with $i \notin \{a, b\}$; such S is losing since it has size $n-3$. Each edge not containing i contributes exactly one such swing, giving $|E| - \deg_G(i)$ swings. - If $|S| = n-2$, then $S \cup \{i\} = N \setminus \{x\}$ for some $x \neq i$. This is winning iff $\deg_G(x) \geq 1$, which holds by (i). The set $S = N \setminus \{x, i\}$ is losing iff $\{x, i\} \notin E$ (otherwise S contains the winning $(n-2)$ -set $N \setminus \{x, i\}$). Hence exactly $(n-1) - \deg_G(i)$ swings of this type. - If $|S| = n-1$, then $S = N \setminus \{i\}$ and $S \cup \{i\} = N$ is winning. But $N \setminus \{i\}$ is losing iff every edge contains i , i.e., G is a star centered at i , excluded by (ii). Hence no swings of this type. - No other cardinality yields a swing.

Summing gives $\beta_i(v_G) = |E| + (n-1) - 2 \cdot \deg_G(i)$. ■

Corollary 2. If G has all vertices but one of common degree d and the exceptional vertex of degree $d + c/2$, and if $|E| + (n-1) - 2d = 2c$, then $Bz(v_G) = \psi^n$. Since $2|E| = nd + c/2$, the existence of such a graph is equivalent to the solvability of $7c = d(2n-8) + 4(n-1)$ (3) in positive integers d, c with $0 \leq d \leq n-2$ and $d + c/2 \leq n-1$, together with the graphicality of the degree sequence $(d, \dots, d, d+c/2)$.

For $6 \leq n \leq 18$, equation (3) admits such a solution exactly for $n \in \{6, 7, 9, 13, 14, 16, 17\}$. For $n \equiv 4 \pmod{7}$ there is no solution at all; for $n \in \{8, 10, 12, 15\}$ the required exceptional degree exceeds $n-1$. Hence the complement-of-edges family alone does **not** cover all n — which is consistent with the conjecture requiring a genuinely different construction for the remaining values.

The condition (3) is solvable for *infinitely many* n : for $n \leq 30$ the solvable values are 6, 7, 9, 13, 14, 16, 17, 20, 21, 23, 24, 27, 28, 30, ... Indeed, the residue classes $n \equiv 6, 0, 2, 3 \pmod{7}$ are solvable for every admissible n with the *constant* common degree $d = 2, 3, 1, 4$ respectively, since then $\deg_G(n) = (4n+4)/7, (5n+7)/7, (3n+1)/7, (6n+10)/7 \leq n-1$. Thus Conjecture 16 holds, by Theorem 1, for **four of the seven residue classes** (i.e. for infinitely many n); the values $n \equiv 1, 4, 5 \pmod{7}$ require a genuinely different construction (given in Sections 4 and 9).

4. The threshold-with-exceptions family

The complement-of-edges family does not cover every n . For the remaining small values a second, equally explicit family is useful. Let E be a set of pairs and F a set of triples of $[n]$. Define the game $v(E, F)$ by $v(S) = 1 \iff |S| \geq n-1$, or $(|S| = n-2 \text{ and } N \notin E)$, or $(|S| = n-3 \text{ and } N \in F)$. (4) That is, the game is “almost” a majority-style threshold at size $n-2$, with a set of *holes* ($N \in E$) removed at level $n-2$ and a set of *exceptions* ($N \in F$) added at level $n-3$.

Lemma (monotonicity). $v(E, F)$ is monotone (hence a simple game) iff no edge of E is contained in a triple of F , i.e. $E \cap \text{Pairs}(F) = \emptyset$, where $\text{Pairs}(F)$ is the set of all two-element subsets of triples in F . Indeed, a winning $(n-3)$ -set $N \setminus X$ ($X \in F$) has a losing $(n-2)$ -superset $N \setminus \{a, b\}$ exactly when $\{a, b\} \in E$ with $\{a, b\} \subseteq X$.

Theorem 2. For any E, F with $E \cap \text{Pairs}(F) = \emptyset$, the swing counts of $v(E, F)$ are $\beta_i(v(E, F)) = C(n-1, 2) + |F| - |E| + 2 \cdot \deg_E(i) - 2 \cdot r_i + e_i$, (5) where $\deg_E(i)$ is the degree of i in the graph E , $r_i = |\{X \in F : i \in X\}|$, and $e_i = |\{X \in F : i \in X, X \setminus \{i\} \in E\}|$.

Proof sketch. Player i swings on a losing S with $S \cup \{i\}$ winning. Enumerate by $|S|$: - $|S| = n-2$ (S losing means $N \notin E$): the unique losing $(n-2)$ -set with $i \notin S$ and $N \in E$ requires $N \ni i$, giving $\deg_E(i)$ swings; - $|S| = n-3$: $S = N \setminus \{i, a, b\}$ with $\{i, a, b\} \notin F$ and $\{a, b\} \notin E$, giving $C(n-1, 2) - |E| + \deg_E(i) - r_i + e_i$ swings; - $|S| = n-4$: $S = N(X \cup \{i\})$ for $X \in F$ with $i \notin X$, giving $|F| - r_i$ swings. Summing yields (5). ■ (The identity is verified exhaustively against direct computation on random instances for $n = 8, 10, 12$.)

Corollary 3. If $\deg_E(i) - r_i$ is constant, say D , for all $i \neq n$ and $\deg_E(n) - r_n = D - c/2$, then $\text{Bz}(v(E, F)) = \psi^n$ with scale $c = (n \cdot C(n-1, 2) + (n-6)|F| - (n-4)|E|)/(2n-1)$ (when this is a positive integer). In particular, such games realize ψ^n for every $n \in \{8, 10, 11, 12, 15\}$ not covered by Corollary 2 (found by solving (5) via mixed-integer programming; see Section 7), while for $n = 18$ a solution was obtained by Kurz and Napel.

5. Explicit games realizing ψ^n

For each $n \in \{6, 7, 9, 13, 14, 16, 17\}$ we give the graph G_n (edges written as pairs) such that $\text{Bz}(v_{G_n}) = \psi^n$; the minimal winning coalitions of v_{G_n} are the complements of these edges. The resulting swing vector is $\beta = c \cdot (2, \dots, 2, 1)$,

displayed as a multiple of c.

n	graph G_n	c	$\beta(v_{G_n})$
6	$C_4(1-2-6-3-1) \cup K_3(4, 5, 6)$	4	(8,8,8,8,4)
7	C_6 on $\{1, \dots, 6\} \cup$ star centered at 7	6	(12, ..., 12, 6)
9	star at 9 on $\{1, 2, 3, 4\} \cup$ matching $\{5, 6\}, \{7, 8\}$	6	(12, ..., 12, 6)
13	star at 13 on $\{1, \dots, 8\} \cup$ matching on $\{1, \dots, 8\} \cup C_4$ on $\{9, \dots, 12\}$	12	(24, ..., 24, 12)
14	C_{13} on $\{1, \dots, 13\} +$ chord $\{6, 10\} \cup$ star at 14 except $\{6, 10\}$	16	(32, ..., 32, 16)
16	star at 16 on $\{1, \dots, 7\} \cup$ matching on $\{8, \dots, 15\}$	12	(24, ..., 24, 12)
17	star at 17 (to all) $\cup C_{16}$ on $\{1, \dots, 16\} \cup$ matching $\{i, i+8\}$ ($1 \leq i \leq 8$)	24	(48, ..., 48, 24)

Each entry is verified directly by computing the swing counts. For instance, the $n = 6$ entry is the game whose minimal winning coalitions are $\{1, 2, 3, 4\}$, $\{1, 2, 3, 5\}$, $\{1, 2, 3, 6\}$, $\{1, 2, 4, 5\}$, $\{1, 3, 4, 5\}$, $\{2, 4, 5, 6\}$, $\{3, 4, 5, 6\}$ (the complements of the seven edges of $C_4 \cup K_3$), whose swing counts are (8,8,8,8,4).

6. A correction to the published example

Kurz and Napel (2014, footnote 19) state that the game on six voters whose minimal winning coalitions are $\{2, 4, 5, 6\}$, $\{2, 3, 4, 5\}$, $\{1, 3, 5, 6\}$, $\{1, 3, 4, 5\}$, $\{1, 2, 4, 6\}$, $\{1, 2, 3, 5\}$ attains the PBI vector $(8,8,8,8,4)/44 = \psi^6$. This is **incorrect**: a direct computation (or a routine check by hand for, e.g., player 6, who has five swings) shows the swing counts are $\beta = (7,7,7,7,9,5)$, $\sum \beta = 42$, so that $Bz(v) = (7,7,7,7,9,5)/42 \neq \psi^6$. The claim that an exact solution exists for $n = 6$ is nevertheless true: the game in Section 5 (seven four-sets) attains ψ^6 . (An exhaustive enumeration of all 7,828,354 simple games on six voters yields 360 games, in two symmetry classes, whose Banzhaf index equals ψ^6 .)

7. Computational verification for $6 \leq n \leq 17$

For the values $n \in \{8, 10, 11, 12, 15\}$ not covered by Corollary 2 we verified existence by formulating the problem as a mixed-integer program: variables $x(S)$

$\in \{0, 1\}$ for the winning status of each coalition S , the monotonicity constraints $x(S) \leq x(T)$ for $S \subseteq T$, the normalization $x(\emptyset) = 0$, $x(N) = 1$, and the swing equations $\sum_{S \not\ni i} (x(S \cup \{i\}) - x(S)) = 2c$ ($i \neq n$), $\sum_{S \not\ni n} (x(S \cup \{n\}) - x(S)) = c$, with $c \geq 1$ integer. Using HiGHS, all these instances are feasible. Table 2 summarizes the scale c found ($n = 6, 7$ by exhaustive enumeration; $n = 13, 14, 16, 17$ by the closed-form construction of Section 4; the five remaining values by mixed-integer linear programming (MILP) — since reproduced at the same scales by the closed-form $\deg_G + r_i = D$ constructions of Corollary 16).

n	6	7	8	9	10	11	12	13	14	15	16	17
c	4	6	8	6	10	14	16	12	16	20	12	24

Note. The scale $c = 20$ for $n = 15$ was found by the CP-SAT model — an improvement over the previously reported $c = 22$; minimality at $n = 15$ is open (a downward sweep at $c = 18$ did not terminate within a 600 s solver budget). Minimality is certified for $n = 8$ ($c = 8$, with $c \in \{2, 4, 6\}$ proven infeasible), $n = 9$ ($c = 6$, $c \in \{2, 4\}$ infeasible) and $n = 10$ ($c = 10$, $c \in \{2, 4, 6, 8\}$ infeasible); see Open problem 3 in Section 11.

For $n = 6$ the existence is established by exhaustive enumeration over all 7,828,354 simple games (360 solutions in two symmetry classes). For $n = 7$, exhaustive enumeration of all games is infeasible; we instead enumerated the 5-uniform family (winning coalitions of size 5) exhaustively and found 490 solutions, all of the complement-of-edges type $C_6 \cup \text{star}_7$. Thus Conjecture 16 holds for every $6 \leq n \leq 17$ by the constructions of Sections 3–5; the value $n = 18$ and all larger values are covered by the closed-form constructions of Section 11 (Corollaries 14–17).

8. Summary of the constructive half

For the inverse voting-design problem, we have delivered the *construction half* in full: an explicit, closed-form simple game whose Penrose–Banzhaf index equals $\psi^n = (2, \dots, 2, 1)/(2n - 1)$ for every $n \geq 6$ — constructively resolving Kurz’s Conjecture 16, whose benchmark status motivated the design study. The main ingredients are:

1. **Two closed-form families** with exact swing formulas (Theorems 1 and 2) — the *complement-of-edges* family (covering $n \equiv 0, 2, 3, 6 \pmod{7}$) and

the *triangle-with-special-triples* family (covering $n \equiv 1, 4, 5 \pmod{7}$) — unified in a single parametric framework, the W-family (Proposition 11), whose swing formula (U) is exact;

2. **A correction of the only published example** (Kurz and Napel 2014, footnote 19);
3. **A complete closed-form construction for every $n \geq 6$** (Corollaries 14, 16, 17), where the five smallest values 8, 10, 11, 12, 15 are handled by the $\deg_G + r_i = D$ structure.

The arithmetic characterization (3) shows that no single complement-of-edges construction can cover all n , and the residue classes differ genuinely; nevertheless, a single parametric framework — the W-family with base degree and a suitable number of (special) triangles — realizes ψ^n for every n .

9. The weighted case: ψ^n is not realizable by any weighted majority game

We now turn to the *obstruction half* of the design decision problem: is the target ψ^n realizable within the weighted majority class — the rule class used by most real voting bodies (parliaments, the EU Council, the IMF, corporate voting)? This is the question a designer must answer before deciding whether to stay in the weighted class. It is precisely Kurz’s **Conjecture 15**, the natural “dual” of Conjecture 16. As stated by Kurz (2015, Conjecture 15) it is quantitative: there exists a universal $c > 0$ with $\|\text{Bz}(v) - \psi^n\|_1 \geq c/n$ for every $n \geq 2$ and every weighted majority game v . In particular no weighted majority game reproduces ψ^n exactly (distance zero is excluded). We certify the exact case for $n \leq 10$, and the quantitative form for $n \leq 8$. Recall that a **weighted majority game** is a simple game $[q; w_1, \dots, w_n]$ with integer weights $1 \leq w_1 \leq \dots \leq w_n$ and quota $q \geq 1$, where S is winning iff $W(S) = \sum_{i \in S} w_i \geq q$. Let $\eta_i = \beta_i([q; w])$ be the swing count. The following identity is a classical structural fact for weighted majority games — a swap/block argument of the type long used in the Banzhaf literature (cf. Felsenthal & Machover 1998). We do not claim it as new; what is new is that it rigidly dissects Kurz’s specific target ψ^n , pinning the obstruction on a subset-sum problem.

Proposition 4 (structural decomposition; swap/block argument). Let $v = [q; w]$ be a weighted majority game with $w_i \geq w_j$. Then $\eta_i - \eta_j = 2 \cdot |\{T \subseteq N \setminus \{i, j\} : q - w_i \leq W(T) < q - w_j\}|$. (K)

Proof. Enumerate the swings of i and j by the subset $T \subseteq N \setminus \{i, j\}$ that excludes both, and by whether the swing coalition S does or does not contain the other player: - i swings on $S = T$ iff $W(T) \in [q - w_i, q)$; - i swings on $S = T \cup \{j\}$ iff $W(T) \in [q - w_i - w_j, q - w_j)$; - j swings on $S = T$ iff $W(T) \in [q - w_j, q)$; - j swings on $S = T \cup \{i\}$ iff $W(T) \in [q - w_i - w_j, q - w_i)$.

Because $w_i \geq w_j$, we have $q - w_i \leq q - w_j$ and $q - w_i - w_j \leq q - w_i$, so $[q - w_j, q) \subseteq [q - w_i, q)$ and $[q - w_i - w_j, q - w_i) \subseteq [q - w_i - w_j, q - w_j)$, and in both inclusions the *removed* interval is the same one, $[q - w_i, q - w_j)$. Summing the four counts over all T and subtracting η_j from η_i cancels the two common pieces and leaves exactly twice the count of T with $W(T) \in [q - w_i, q - w_j)$. ■

The identity was verified on 12,000 random weighted games ($n = 3 \dots 12$, weights ≤ 30) with 0 failures (`probe6_identity.py`).

Corollary 5 (parity; resolves Observation 7 in the Appendix). If $Bz(v) = \psi^n$, then the scale factor c is even. Two independent proofs: - (*any simple game*, Dubey–Shapley 1979): in every simple game all swing counts have a common parity. Here $\beta = (2c, \dots, 2c, c)$; since $2c$ is even, the common parity is even, so c is even. - (*weighted, from (K)*): if w_s is the special player ($\eta_s = c$), then for any $i \neq s$, $c = \eta_i - \eta_s = 2 \cdot |\{T : q - w_i \leq W(T) < q - w_s\}|$, so c is even.

Corollary 6 (the special player has strictly minimal weight). Suppose η_s is the unique minimum swing count and $\eta_i > \eta_s$ for every $i \neq s$. Then $w_s < w_i$ for all $i \neq s$. Indeed $\eta_i - \eta_s > 0$ forces the interval $[q - w_i, q - w_s)$ to contain some subset sum, so it is nonempty and $w_i > w_s$. Thus, in any weighted game whose Banzhaf index is proportional to ψ^n , the singled-out voter must be the *unique lightest* voter — an a-priori monotonicity constraint on the weights.

Corollary 7 (interval decomposition and the mirror obstruction). With s the special player and $\eta = (2c, \dots, 2c, c)$, identity (K) applied to the pair (i, s) yields, for every $i \neq s$, $|\{T \subseteq N \setminus \{i, s\} : q - w_i \leq W(T) < q - w_s\}| = c/2$, (L) so the excess $c/2$ of each non-special voter over the special voter is realized by subset sums of $N \setminus \{i, s\}$ inside the weight gap $[q - w_i, q - w_s)$. Now fix two non-special voters i, j with $w_i \geq w_j$ (WLOG) and split the window count in (L) at (i, s) according to whether j lies in T , and likewise the count at (j, s) according to whether i lies in T : $|\{T \subseteq N \setminus \{i, s\} : q - w_i \leq W(T) < q - w_s\}| = |\{T \subseteq N \setminus \{i, j, s\} : q - w_i \leq W(T) < q - w_s\}| + |\{T \subseteq N \setminus \{i, j, s\} : q - w_i - w_j \leq W(T) < q - w_s - w_j\}|, |\{T \subseteq N \setminus \{j, s\}$

$: q - w_j \leq W(T) < q - w_s\} = |\{T \subseteq N \setminus \{i, j, s\} : q - w_j \leq W(T) < q - w_s\}| + |\{T \subseteq N \setminus \{i, j, s\} : q - w_i - w_j \leq W(T) < q - w_s - w_i\}|$, both equal to $c/2$. Since $w_i \geq w_j > w_s$ (Corollary 6), the relevant weight windows nest as disjoint unions $[q - w_i, q - w_s) = [q - w_i, q - w_j) \sqcup [q - w_j, q - w_s)$, $[q - w_i - w_j, q - w_s - w_j) = [q - w_i - w_j, q - w_s - w_i) \sqcup [q - w_s - w_i, q - w_s - w_j)$, so subtracting the two equalities leaves the **mirror identity** $|\{T \subseteq N \setminus \{i, j, s\} : q - w_i \leq W(T) < q - w_j\}| + |\{T \subseteq N \setminus \{i, j, s\} : q - w_s - w_i \leq W(T) < q - w_s - w_j\}| = 0$. (M) Both summands are nonnegative, hence they vanish individually: $|\{T \subseteq N \setminus \{i, j, s\} : q - w_i \leq W(T) < q - w_j\}| = 0$ and $|\{T \subseteq N \setminus \{i, j, s\} : q - w_s - w_i \leq W(T) < q - w_s - w_j\}| = 0$. (M') In words: for every pair of non-special voters i, j with $w_i \geq w_j$, the weight-gap window $[q - w_i, q - w_j)$ — and its translate by w_s — is *sum-free* for the profile of the remaining $n - 3$ voters. Together with (L) this is a sharp rigidity statement: the $c/2$ sums realizing voter i 's excess over s must come exclusively from the part $[q - w_i - w_j, q - w_s - w_j)$ of the gap window, never from the sub-window $[q - w_i, q - w_j)$ itself; and the reflected sub-window $[q - w_s - w_i, q - w_s - w_j)$ is likewise empty. (Verified numerically: across 2.28×10^5 random weighted games with $\eta_i = \eta_j$, both windows in (M') were empty in every case.)

The Banzhaf-weighted obstruction property (M'). We record the rigidity statement just derived as the paper's central structural obstruction. Under the exact hypothesis $\eta = (2c, \dots, 2c, c)$, *any* realizing game must satisfy (L) and (M') — they are necessary conditions. They are also so tight that the full constraint system — (L), (M'), together with the exact swing profile — is *provably infeasible*, as certified by CP-SAT at the classical weight bound $w_i \leq 2^{n-1}$ for $n = 6, \dots, 10$ (Section 9, below). In this sense (M') is the structural certificate of the obstruction: the subset-sum windows it forces to be empty are precisely the degrees of freedom whose infeasibility the deterministic solver establishes. The property is not merely cosmetic: the mirror constraints of (M') are the pruning constraints of the enhanced feasibility model (`conj15_feas2.py`), which decides $n = 6$ in 2.5 s (vs 2.9 s) and $n = 7$ in 19 s — about six times faster than the unpruned model at $n = 7$ (117 s) — and are thus what makes the $n \geq 9$ frontier approachable at all. We are not aware of this reflected sum-freeness rigidity in the threshold-logic or voting-power literature; the closest structural studies — Chow's (1961) theorem on Chow parameters (a recovery problem) and the inverse-power-index certification results of Alon & Edelman (2010) and Kurz (2016) — address reconstruction and worst-case

approximation, not the exact-hypothesis rigidity used here.

Two-tier generalization $\psi^n(a, b)$ and a family-level no-go. For $a = b$ the target $\psi^n(a, b) = (a, \dots, a, b) / ((n-1)a + b)$ equals ψ^n with c scaled. For $a > b$ the W-family triangle construction of Proposition 11 realizes $\psi^n(a, b)$ with special degree $s = \lfloor (a-b)(5n-5-p) - 4aq + 12b \rfloor / (3a+b)$, (N) where p pure and q special triangles are used. Systematic search (probe4/probe5) gives: - **(3,1) is uniform:** for all $n \in [6, 80]$, $s = n-1$ with $p = 0$, $q = 1$ realizes $\psi^n(3, 1)$ — the construction $G = \text{star}(n) \cup C_{n-1} \cup \{1, 2\}$, $F = \{\text{triangle on the special player}\}$ works for every n , no case distinction. - **(k,1), $k \geq 4$ is obstructed within this family:** for $k \geq 4$ the required degree $s \approx 5(k-1)n/(3k+1)$ exceeds $n-1$ for large n , so no triangle construction of this shape exists. Coverage rates over $n \in [6, 80]$: $k=2$: 73/75 (gaps 8, 11); $k=3$: 75/75; $k=4$: 40/75; $k=5$: 28/75; $k=6$: 9/75; $k=7$: 8/75. Combined with Corollaries 6–7, this is strong evidence that $\psi^n(k, 1)$ for $k \geq 4$ is not weighted-realizable at all; a complete proof is the subject of ongoing work (Corollary 6 pins w_s minimal; (L) and (M) then over-constrain the subset-sum profile).

Computational obstruction (exact). For ψ^n itself, exact feasibility of a weighted realization with swing vector $(2c, \dots, 2c, c)$ was settled by CP-SAT (`conj15_feas.py`). Infeasibility is certified at the classical weight bound $W = 2^{n-1}$: every weighted majority game on n voters has an integer representation with all weights in $[1, 2^{n-1}]$ (a standard bound, implied for $n \geq 3$ already by Muroga’s sharper minimal-weight bound $3^{n/3}$), so infeasibility at $W = 2^{n-1}$ is conclusive:

	n	$W = 2^{n-1}$	model	verdict	runtime
6	32		standard	INFEASIBLE	2.9 s
7	64		standard	INFEASIBLE	117 s
8	128		+symmetry breaking	INFEASIBLE	316 s
9	256		enhanced (<code>feas2</code>)	INFEASIBLE	1562.9 s
10	512		enhanced (<code>feas2</code>)	INFEASIBLE	9527.4 s

The pipeline is three-tiered, and we are explicit about which tier backs each claim. **Tier 1 (exhaustive):** $n = 6$ is settled by enumerating all 1,111 weighted games, guaranteed. **Tier 2 (deterministic):** $n = 7, 8, 9$ and 10 are settled by CP-SAT at the conclusive weight bound $W = 2^{n-1}$ (standard model for $n = 7$,

8; the enhanced model below for $n = 9, 10$), guaranteed. **Tier 3 (budgeted):** $n \geq 11$ runs the same solver under a time budget, so a negative outcome there is heuristic evidence for a trend, not a proof, and is labeled accordingly.

$n = 6$ agrees with the exhaustive enumeration of all 1,111 weighted games on 6 voters (`weighted_enum.py`), whose best L_1 – distance to ψ^6 is $5/44 \approx 0.1136$; hence no weighted game attains distance 0. The $n = 8$ run required the symmetry-breaking constraint that non-special voters can be relabeled in non-decreasing weight order (all have equal target swing 2c); with it the infeasibility certificate is found in 316 s, whereas the unconstrained model is inconclusive after 20 min. $n = 9$ is substantially harder: the standard model exhausts a full 2 h budget without a certificate (UNKNOWN), a dramatic jump consistent with the exponential growth of the subset-sum constraint structure. The enhanced model (`conj15_feas2.py`), which additionally enforces the sharp mirror constraints (M') — valid since they are consequences of the exact hypothesis — then decides $n = 9$ in 1562.9 s (3584 mirror constraints): **INFEASIBLE**. At $n = 10$ the same enhanced model decides **INFEASIBLE** in 9527.4 s (9216 mirror constraints), while the standard model again exhausts a 2 h budget without a certificate (UNKNOWN). This closes the exact obstruction for all $n \leq 10$, the strongest conclusive statement currently available; the enhanced model already gives a $6\times$ speedup on the smaller scales ($n = 6$: 2.5 s, $n = 7$: 19 s), and its growing mirror-constraint set is what makes the $n \geq 9$ frontier approachable at all. This is exactly Kurz’s Conjecture 15, which asserts the quantitative statement that for some universal $c > 0$, $\|Bz(v) - \psi^n\|_1 \geq c/n$ for *every* weighted majority game v and every $n \geq 2$. The exact obstruction for $n = 6, \dots, 10$ is the first step; the constant c is extracted from the optimal L_1 errors computed just below.

Optimal L_1 – distance (exact computation, `conj15_11.py`). For each n we compute the exact minimal value of $\|Bz(v) - \psi^n\|_1$ over all weighted majority games, by fixing the total number of swings $T = \sum \eta_i$ and minimizing the integer objective $\sum_{i \neq s} |(2n-1)\eta_i - 2T| + |(2n-1)\eta_s - T|$ in CP-SAT, enumerating T over its full range and using the same weight bound $W = 2^{n-1}$; the special player is fixed at position n without loss (relabeling). The method reproduces the exhaustive $n = 6$ optimum exactly (`conj15_11.py n=6 T ≤ 45`) : $L_1 = 5/44 \approx 0.113636$ at **$T = 32$, with weights $w = (3,1,3,3,3,1)$, quota $q = 11$** , matching the brute-force enumeration of all 1,111 weighted games — a strong check of both methods.

	n	optimal L_1	T	$n \cdot L_1$	1/n
6	5/44	≈ 0.113636	32	≈ 0.682	0.167
7		≈ 0.085470	45	≈ 0.598	0.143
8		≈ 0.066667	60	≈ 0.533	0.125

(The $n = 8$ value is certified as the global optimum: $T = 60$ is OPTIMAL by CP-SAT, and the total swing count was swept over its full range to the theoretical maximum $8 \cdot 2^7 = 1024$ — `conj15_11_range.py` — whose best value outside $T = 60$ is $L_1 = 0.116667$ at $T = 280$, well above the incumbent.) For $n = 7$ the optimal profile is $(5, 7, 7, 7, 7, 7, 5)$ with weights $(2, 64, 64, 64, 64, 64, 1)$ and quota 65; the value is certified OPTIMAL by CP-SAT (re-verified with a 600 s budget), and the full sweep over $T \leq 448$ — the theoretical maximum $7 \cdot 2^6$ of the total swing count — confirms that no other total swing count improves on it: the best value in the complementary range $T \geq 121$ is $L_1 = 0.119535$ at $T = 139$ (`conj15_11_range.py`), well above the incumbent. For $n = 8$ the optimum is attained at profile $(6, 8, 8, 8, 8, 8, 8, 6)$ with weights $(2, 3, 3, 3, 3, 3, 3, 1)$ and quota 4. The column $n \cdot L_1$ is the empirical anchor for the universal constant in Conjecture 15: since $\|\cdot\|_1 \geq c/n$, any admissible c is at most $\inf_n n \cdot L_1(n)$; the values above give $c \leq \min(30/44, 7 \cdot 0.08547, 8 \cdot 0.066667) \approx 0.533$. We stress the distinction between proof and certification, which is central to how the computational results below are to be read: the value $5/44$ for $n = 6$ is an exhaustive optimum, and 0.085470 for $n = 7$ is a *certified* optimum (over the full range $T \leq 448$ of the total swing count); the value 0.066667 for $n = 8$ is likewise a *certified* optimum, the total swing count having been swept over its full range to the theoretical maximum $T = 8 \cdot 2^7 = 1024$. The exact target $(2c, \dots, 2c, c)$ is infeasible for $6 \leq n \leq 10$ by the computation above, so the L_1 — *optimum* is strictly positive. It is the monotone decrease $n \cdot L_1 = 0.682 \rightarrow 0.598 \rightarrow 0.533$ that supports the *conjectured* $\Theta(1/n)$ asymptotics for all n ; deriving an analytic lower bound $\|Bz(v) - \psi^n\|_1 \geq c/n$ with an effective constant — not merely certification on the first three scales — remains open. The optimal profiles are asymmetric: the two lightest voters — the special voter and one further voter — carry the reduced swing count, and they do not sit at the ideal scale $T = (2n - 1)c$ ($n = 6$: profile $(6, 4, 6, 6, 6, 4)$ at weights $(3, 1, 3, 3, 3, 1)$, quota 11; $n = 7$: profile $(5, 7, 7, 7, 7, 7, 5)$ at weights $(2, 64, 64, 64, 64, 64, 1)$, quota 65; $n = 8$: profile $(6, 8, 8, 8, 8, 8, 8, 6)$ at weights $(2, 3, 3, 3, 3, 3, 3, 1)$, quota 4). This is consistent with

Corollary 6: in the exact case the special voter alone would be uniquely lightest, whereas near-optimal games distribute the reduction between the two lightest voters.

10. Computational methodology: exact construction and certified obstruction

This section collects the two computational engines behind the paper as a reusable methodology — with the complexity statements and instance tables expected of an operations-research study. Every claim below is backed by a named script and a full log in the companion code package (Section 11).

10.1 Constructive engine: closed forms and the W-family MILP

Task. Given n , produce an explicit simple game with $Bz(v) = \psi^n$, or show that the family under consideration cannot cover n .

1. **Residue arithmetic.** For the complement-of-edges family, realization reduces (Theorem 1) to the diophantine equation $7c = d(2n-8) + 4(n-1)$ in positive integers d, c with $1 \leq d \leq n-2$, $d + c/2 \leq n-1$, together with Erdős–Gallai graphicality of the degree sequence $(d, \dots, d, d+c/2)$. This is an $O(1)$ decision per n , and the family covers exactly the four residue classes $n \equiv 0, 2, 3, 6 \pmod{7}$ (Proposition 8).
2. **W-family MILP/CP-SAT.** For the remaining residue classes, realization over the W-family of Proposition 11 is formulated as a MILP with edge variables x_e ($e \in G$) and triple variables y_T ($T \in F$), the bilinear products $z = y \cdot x$ linearized in the standard way, and the exact swing formula (U) together with the monotonicity constraint $\text{Pairs}(F) \subseteq G$ as linear constraints. **Model size:** $O(n^3)$ **variables and constraints** ($C(n,2) + C(n,3)$ variables) — versus 2^n for the general monotone Boolean MILP of Section 7, which becomes intractable near $n = 18$. The compressed model solves every instance up to $n = 64$ in seconds to minutes (instance data below); its practical limiting factor is memory, reached around $n \approx 80$.
3. **Closed forms.** Proposition 13 gives explicit triangle constructions for the three hard residue classes $n \equiv 1, 4, 5 \pmod{7}$, valid for all $n \geq 22, 18, 19$ respectively; the small values 8, 10, 11, 12, 15 are handled by the $\deg_G + r_i = D$ structure (Corollary 16). Combined with Step 1 and Corollary 14, **every $n \geq 6$ admits an explicit closed form** (Corollary 17).

Representative W-family instances (hard residue classes; c = scale factor). All are explicit closed forms, re-verified by direct swing enumeration against ψ^n (full data in appendix-data-n80.json):

	n	n mod 7	c	edges	triples	construction
	18	4	24	33	6	$\text{deg}+r = D = 4$
	22	1	26	32	5	$\text{tri}(D3, f5, q3)$
	25	4	30	36	6	$\text{tri}(D3, f6, q3)$
	39	4	50	62	6	$\text{tri}(D3, f6, q2)$
	67	4	90	114	6	$\text{tri}(D3, f6, q0)$
	71	1	96	123	5	$\text{tri}(D3, f5, q0)$
	74	4	100	127	6	$\text{tri}(D3, f6, q0)$
	78	1	106	136	5	$\text{tri}(D3, f5, q0)$

Here $\text{tri}(D, f, q)$ denotes the triangle construction of Proposition 13 with base degree D , f disjoint triangles and q special triples (containing the special player), and $\text{deg}+r = D$ the small-value structure of Corollary 16. The structural insight for the hard residues is that the solutions use few exceptional triples — $f \in \{4, 5, 6\}$ — and that for $n \geq 67$ the special player need not participate in any special triple ($q = 0$; Corollary 15).

10.2 Verification engine: certification of weighted obstructions

Task. Given n , decide whether any weighted majority game $[q; w]$ on n voters realizes the exact target swing vector $(2c, \dots, 2c, c)$; if none exists, produce a certificate.

1. **Constraint system and conclusive weight bound.** The model (`conj15_feas.py`) has variables $w_1 \leq \dots \leq w_n \leq W$, q , and the swing counts η_i , together with the exact profile equations ($\eta_i = 2c$ for $i \neq s$, $\eta_s = c$) and the structural consequences (L) , (M') of Section 9. Infeasibility at $W = 2^{n-1}$ is conclusive, because every weighted majority game on n voters admits an integer representation with all weights in $[1, 2^{n-1}]$ (Muroga’s sharper bound $3^{n/3}$ applies for $n \geq 3$). A solver certificate of infeasibility at this bound is therefore a proof for that n , not a heuristic.
2. **Symmetry breaking.** Non-special voters may be relabeled in nondecreasing weight order (they carry equal target swing $2c$). Without this

constraint $n = 8$ remains inconclusive after 20 min; with it the certificate is found in 316 s.

3. **Mirror pruning.** The sharp constraints (M') of Corollary 7 are consequences of the exact hypothesis, hence valid; enforcing them (`conj15_feas2.py`) yields a roughly six-fold speedup ($n = 6$: 2.5 s vs 2.9 s; $n = 7$: 19 s vs 117 s, a six-fold speedup) and — decisively — makes the $n \geq 9$ frontier decidable at all.

Certified obstructions (deterministic, at $W = 2^{n-1}$) :

	n	$W = 2^{n-1}$	model	verdict	runtime
6	32		standard	INFEASIBLE	2.9 s
7	64		standard	INFEASIBLE	117 s
8	128		+symmetry breaking	INFEASIBLE	316 s
9	256		+mirror	INFEASIBLE	1562.9 s
10	512		+mirror	INFEASIBLE	9527.4 s

The $n = 6$ verdict agrees with exhaustive enumeration of all 1,111 weighted games (Tier 1); $n = 7$ –10 are Tier-2 deterministic certificates. Under a two-hour budget the standard (unpruned) model remains UNKNOWN at $n = 9$ and $n = 10$ — the mirror constraints are what closes these cases. For $n \geq 11$ the same solver is run budgeted, so a negative outcome there is heuristic (Tier 3) and is labelled as such. The mirror constraints — 3584 at $n = 9$ and 9216 at $n = 10$, *i.e.* $C(n-1, 2) \cdot 2^{n-2}$ window conditions, one per subset of the $n-3$ remaining voters and for both windows — are themselves a measure of the rigidity that (M') forces on any realizing game.

10.3 Exact optimization over total swing count

For the quantitative form of Conjecture 15 we compute the minimal L_1 – distance $\|Bz(v) - \psi^n\|_1$ exactly. Fixing the total number of swings $T = \sum \eta_i$, the integer objective $\sum_{i \neq s} |(2n-1)\eta_i - 2T| + |(2n-1)\eta_s - T|$ is minimized in CP-SAT; T is enumerated over its full range up to the theoretical maximum $n \cdot 2^{n-1}$, at the same conclusive weight bound (`conj15_l1.py`, `conj15_l1_range.py`). The method reproduces the exhaustive $n = 6$ optimum 5/44 (weights (3,1,3,3,3,1), quota 11) exactly — a strong cross-check of both methods. Certified optima over the full swing range: $n = 7$, $L_1 \approx 0.085470$ at

$T = 45$; $n = 8$, $L_1 \approx 0.066667$ at $T = 60$ (over $T \leq 1024$). These give $c \leq \min(30/44, 7 \cdot 0.08547, 8 \cdot 0.066667) \approx 0.533$. At $n = 9$ the problem hits the CP-SAT complexity cliff: with 120 s per T-value, 23 of the 112 values in $[9, 120]$ are decided, 0 are certified infeasible, and the best found $L_1 \approx 0.1042$ (at $T = 70$) is not certified. We report this honestly as the current frontier: **$c \leq 0.533$ is certified for $n \leq 8$; no certified value exists at $n = 9$.**

10.4 Reproducibility

Every claim above is tied to a named script — `conj15_feas.py`, `conj15_feas2.py`, `conj15_l1.py`, `conj15_l1_range.py`, `weighted_enum.py`, `probe6_identity.py`, and the W-family MILP/CP-SAT solver of Proposition 12 — with full logs in the companion package; solver versions, hardware, and per-run parameters are recorded in the package README.

11. The W-family: unified constructions, structural properties, and open problems

This section develops the W-family into its full scope: it proves the unified swing formula that the construction half (Section 8) relies on, establishes the structural properties that govern which residue classes admit closed-form realizations, and collects the open problems that remain. All results are labelled with their derivation or verification method; unproven numerical observations are explicitly marked as such. Reproducible code is given in the appendix.

Proposition 8 (Exact description of complement-of-edges coverage; Erdős–Gallai computation)

For each $n \geq 6$, the complement-of-edges family (Theorem 1) is solvable at n if and only if there are positive integers d, c with $7c = d(2n - 8) + 4(n - 1)$, $1 \leq d \leq n - 2$, c even, $d + c/2 \leq n - 1$, $(n \cdot d + c/2)$ even, and the degree sequence $(d, \dots, d, d + c/2)$ passes the Erdős–Gallai test and is constructible by Havel–Hakimi.

Lemma A (two-value Erdős–Gallai reduction). Let $1 \leq d < D \leq n - 1$. The degree sequence $(d, \dots, d, D)$ with $n - 1$ copies of d is graphical if and only if $(n - 1)d + D$ is even, $D \leq n - 1$, and $D \leq d(n - d)$.

Proof. Sorting descending as $(D, d, \dots, d)$, the Erdős–Gallai inequalities are,

for every k , $\sum_{i \leq k} a_i \leq k(k-1) + \sum_{i > k} \min(a_i, k)$. For $k = 1$ this reads $D \leq n-1$. For $2 \leq k \leq d$ every $a_i = d \geq k$, so the RHS is $k(k-1) + (n-k)k = k(n-1)$; the strongest such inequality is at $k = 2$, i.e. $D \leq 2(n-1) - d$, which is implied by $D \leq n-1$. For $k > d$ the RHS is $k(k-1) + (n-k)d$, and the strongest case is $k = d+1$, reading $D \leq d(d+1) + (n-d-1)d = d(n-d)$. Hence, besides parity and $D \leq n-1$, the only content is $D \leq d(n-d)$. (Sufficiency is the Erdős–Gallai theorem.) ■

Analytic conclusion. Applying Lemma A to the four residue classes with the constant d above ($n \equiv 6, 0, 2, 3 \pmod{7}$; $D = d + c/2 = (4n+4)/7, (5n+7)/7, (3n+1)/7, (6n+10)/7$), the conditions hold for every admissible n : for $n \equiv 6$ ($d = 2$) both $D \leq n-1$ and $D \leq 2(n-2)$ hold for all $n \geq 6$; for $n \equiv 0$ ($d = 3$) $D \leq 3(n-3)$ always and $D \leq n-1$ for $n \geq 7$; for $n \equiv 2$ ($d = 1$) both hold for all $n \geq 9$; for $n \equiv 3$ ($d = 4$) $D \leq 4(n-4)$ always and $D \leq n-1$ for $n \geq 17$. Parity: $(n-1)d + D$ equals $(18m+14), (26m-2), (10m+2), (34m+12)$ respectively for $n = 7m+r$, all even. Hence the complement-of-edges family covers $n \equiv 6 \pmod{7}$ from $n \geq 6$, $n \equiv 0$ from $n \geq 7$, $n \equiv 2$ from $n \geq 9$, and $n \equiv 3$ from $n \geq 17$ — with no computational verification. (The value $n = 10$, the only $n \equiv 3$ with $6 \leq n < 17$, is covered by Corollary 16.)

Corollary 9 (new infinite families). All admissible n in the following four residue classes are realized by complement-of-edges constructions: $n \equiv 6 \pmod{7}$, $d = 2$; $n \equiv 0 \pmod{7}$, $d = 3$; $n \equiv 2 \pmod{7}$, $d = 1$; $n \equiv 3 \pmod{7}$, $d = 4$. Each construction is a complement-of-edges game whose graph is “a star centered at the special player plus a structured graph”; explicit graphs for $n = 20, 21, 23, 24$ etc. were built and independently re-checked by both the swing formula and (for small n) exhaustive enumeration. The complement-of-edges family therefore covers **4/7 of all integers**, far beyond the two residue classes of the main text.

Uncovered residue classes: $n \equiv 1, 4, 5 \pmod{7}$ ($n = 18, 19, 22, 25, 26, \dots$ in the main text).

Necessary property: all $(n-1)$ -sets are winning

Proposition 10 (necessary; computational verification + proof sketch). If the Banzhaf index of a simple game v equals ψ^n , then every $(n-1)$ -element coalition is winning.

Verification: exhaustive enumeration of all 7,828,354 simple games on 6 voters

(all 360 solutions satisfy it); for $n = 7, 8$ the MILP becomes infeasible when the constraint “some $(n - 1) - \text{set}$ is losing” is added.

Proof sketch: if $N \setminus \{x\}$ is losing, player x swings on $S = N \setminus \{x\}$; moreover, on each $(n - 2) - \text{set}$ $S = N \setminus \{x, j\}$, player x may swing (because $N \setminus \{j\}$ is winning) while player j never does (because $N \setminus \{x\}$ is losing), so x accumulates extra swings at the top levels, contradicting $\beta \propto (2, \dots, 2, 1)$. A complete proof needs a level-by-level count; left as an unfinished proof in the work record.

Use: it shrinks the search space to games in which all $(n - 1) - \text{sets}$ win (adopted in Task F).

A unified framework: the W-family

Proposition 11 (unifying theorem). For a graph G and a family F of triples with $\text{Pairs}(F) \subseteq G$ (monotonicity), define $v(G, F)(S) = 1 \iff |S| \geq n - 1$, or $(|S| = n - 2 \text{ and } N \notin G)$, or $(|S| = n - 3 \text{ and } N \notin F)$. Then $v(G, F)$ is a simple game and $\beta_i(v(G, F)) = |G| + (n - 1) + |F| - 2 \cdot \deg_G(i) - 2 \cdot r_i + e_i$, (U) where $r_i = |\{X \in F : i \in X\}|$ and $e_i = |\{X \in F : i \in X, X \setminus \{i\} \notin G\}|$.

- With $G = E$ and $F = \emptyset$, (U) reduces to Theorem 1 (complement-of-edges);
- With $G = \text{complement}(E)$ and arbitrary F , (U) reduces to Theorem 2 (threshold-with-exceptions).

Verification: for $n = 6..10$, (U) agrees with direct swing counts on random monotone (G, F) instances. This unification shows that the two construction families belong to one parametric framework, and provides a single equation for finding new constructions.

Proposition 12 (Task F: an $O(n^3) - \text{variable}$ MILP over the W-family; verification extended to $n \geq 18$)

The general MILP of Section 7 has 2^n variables and becomes infeasible around $n = 18$. Using Proposition 11’s W-family, we compress the search to variables x_e (edge $e \in G$, $C(n, 2)$ of them) and y_T (triple $T \in F$, $C(n, 3)$ of them), with the bilinear terms $z = y \cdot x$ linearized in the standard way, yielding a MILP with $O(n^3)$ **variables** (formula (U) and monotonicity $\text{Pairs}(F) \subseteq G$ as linear constraints). This MILP solves all instances up to $n \leq 64$ in seconds to minutes; every resulting construction is an explicit closed form (Corollaries 14–17, Proposition 13), re-verified by formula (U). The scales c are:

{ % do not increment counter

n	n mod 7	c	construction
18	4	24	deg+r=D
19	5	22	tri(D3,f4,q2)
20	6	20	comp-edge
21	0	26	comp-edge
22	1	26	tri(D3,f5,q3)
23	2	18	comp-edge
24	3	36	comp-edge

}

{ % do not increment counter

n	n mod 7	c	construction
25	4	30	tri(D3,f6,q3)
26	5	32	tri(D3,f4,q2)
27	6	28	comp-edge
28	0	36	comp-edge
29	1	36	tri(D3,f5,q2)
30	2	24	comp-edge
31	3	48	comp-edge

}

Here $\text{tri}(D, f, q)$ and comp-edge are as in Proposition 13 / Section 10.1.

Then $n = 32 (\equiv 4) c=40$, $n=33 (\equiv 5) c=42$, $n=34 (\equiv 6) c=36$, $n=35 (\equiv 0) c=46$, $n=36 (\equiv 1) c=46$; $n=37 (\equiv 2) c=30$ and $n=38 (\equiv 3) c=60$ by complement-of-edges; $n=39 (\equiv 4) c=50$. Further: $40(\equiv 5) c=52$, $43(\equiv 1) c=56$, $46(\equiv 4) c=60$, $47(\equiv 5) c=62$, $50(\equiv 1) c=66$, $53(\equiv 4) c=70$, $54(\equiv 5) c=72$, $57(\equiv 1) c=76$, $60(\equiv 4) c=80$, $61(\equiv 5) c=82$, $64(\equiv 1) c=86$, all rechecked by (U).

Closed forms cover every $n \geq 67$ (supplement to Proposition 12).

Corollary 15 shows that for $n \geq 67$ the special player need not participate in any F-triple ($q = 0$). Applying the triangle construction directly yields the

remaining values: $n = 67 \pmod{4}$ $c = 90$, $n = 68 \pmod{5}$ $c = 92$, $n = 71 \pmod{1}$ $c = 96$, $n = 74 \pmod{4}$ $c = 100$, $n = 75 \pmod{5}$ $c = 102$, $n = 78 \pmod{1}$ $c = 106$, with $|G| = 123$, $|F| = 5$ at $n = 71$ and $|G| = 136$, $|F| = 5$ at $n = 78$. **Hence every n with $6 \leq n \leq 80$ is covered without gaps, for all residue classes mod 7** (verified: 75/75 values in $[6, 80]$). The structural insight for the hard residues is that their solutions use few exceptional triples: the triangle construction needs at most six ($|F| \in \{4, 5, 6\}$), and for $n \geq 67$ none of them involves the special player. For $n \geq 81$ the $O(n^3)$ model exhausts memory around $n \approx 80$, recorded as the second computational lower bound of this method.

Methodological remark: the speed of the W-family MILP/CP-SAT ($n = 18$ in about 5 s) shows that the unified framework of Proposition 11 not only unifies the constructions but also gives a scalable computational route; this is a direct payoff of Tasks C/E.

Closed-form constructions for the hard residue classes; full resolution

Proposition 13 (found by exploration, verified via the proven formula (U)). For each of the three residue classes $n \equiv 1, 4, 5 \pmod{7}$, there is an explicit closed-form construction realizing ψ^n , valid for all n from a small threshold:

residue	f	c	special degree s	valid for
$n \equiv 1$	5	$(10(n-1) - 28)/7$	$(c+6)/2 - q$	$n \geq 22$
$n \equiv 4$	6	$(10(n-1) - 30)/7$ for $n \geq 25$ ($D = 3$); $c = 24$ at $n = 18$ ($D = 4$)	$D + c/2 - q$	$n \geq 18$ ($D = 4$ at $n = 18$, else $D = 3$)
$n \equiv 5$	4	$(10(n-1) - 26)/7$	$(c+6)/2 - q$	$n \geq 19$

Here $q \in \{0, \dots, f\}$ is the number of *special triples* — F-triples containing the special player (equal to the number of triples in which the special player participates). The construction takes f disjoint triangles: q of them of the form $\{n, a, b\}$ (special triple) and $p = f - q$ of them involving only non-special players; every player in a triangle has degree 2 in G (base degree $D = 3$, so each triangle

player has degree $D - 1 = 2$) and lies in exactly one F-triple ($r_i = 1, e_i = 0$). The remaining $n - 1 - 3f + q$ non-special players have degree $D = 3$ and $r_i = 0$; the special player has degree $s = D + c/2 - q$ (for $D = 3$ this is $(c+6)/2 - q$). By formula (U) the swing counts are $\beta_i = |G| + (n - 1) + f - 6 = 2c$ (non-special), $\beta_n = |G| + (n - 1) + f - 2s - 2q = c$. The smallest feasible q admits the closed form $q = \max(0, \lceil (K - n)/14 \rceil)$, $K = 57$ ($n \equiv 1$), 67 ($n \equiv 4$), 47 ($n \equiv 5$). (6) Indeed, writing $\delta = 2, 1, 3$ for $n \equiv 1, 4, 5$ respectively, direct substitution ($D = 3, c = (10(n - 1) - A)/7$ with $A = 28, 30, 26$) gives the special degree $s = (5n + \delta)/7 - q$ and the residual degree sequence $(3^a, 2^x)$ on the $m = n - 1 - 3f + q$ non-triangle players, where $x = s - 2q$ and $a = m - x = (2(n - K) + 28q)/7$. Since $n \equiv K \pmod{7}$, a is **always even**. The feasibility conditions $x \geq 0, a \geq 0, a \leq (m+1)/2, q \leq f, m \geq 4, s \leq n - 1$ all hold at (6): $x \geq 0 \iff q \leq (5n + \delta)/21$ and $a \leq (m+1)/2 \iff 49q \leq 3n + 4K - 21f$ are the binding ones, each holding for every $n \geq n_0$ once the elementary inequalities are checked (e.g. $n \geq (3K + 42 - 2\delta)/13 < n_0$ for $x \geq 0$; $n \geq (3.5K + 49 - (4K - 21f))/6.5 < n_0$ for the chord condition); $a \geq 0$ is built into (6), $q \leq f$ and $s \leq n - 1$ are automatic. The residual graph $(3^a, 2^x)$ is then constructed explicitly: place the a degree-3 vertices at the even positions $0, 2, \dots, 2a - 2$, add the cycle $(0 \ 1 \ \dots \ m - 1 \ 0)$, and add the matching chords $(2j, 2j+2)$ for $j = 0, 2, \dots, a - 2$. Each even-position vertex is the endpoint of exactly one chord, so it has degree 3; all other vertices have degree 2. ($a \leq (m+1)/2 \iff 2a - 2 \leq m - 1$ makes the positions exist; a even closes the chord set.) Thus (6) realizes ψ^n for every $n \equiv 1, 4, 5 \pmod{7}$ with $n \geq 22, 25, 19$ respectively, with no search and no reliance on numerical verification.

Corollary 14 (every $n \geq 18$ has a closed form). Since $n \equiv 0, 2, 3, 6 \pmod{7}$ are covered by the complement-of-edges construction of Proposition 8 (by Lemma A, analytically for every admissible n) and $n \equiv 1, 4, 5$ by the general triangle construction of Proposition 13 (by the closed form (6), analytically for $n \geq 22, 25, 19$ respectively), **every $n \geq 18$ admits an explicit closed-form simple game with $\mathbf{Bz}(\mathbf{v}) = \psi^n$** . For $n = 18$ this uses base degree $D = 4$ with six disjoint triangles, three of which contain the special player: $c = 24, |G| = 33, |F| = 6$.

Corollary 15 (unified parametric family). With base degree $d = 3$ for all non-special players and $f =$ the smallest non-negative integer with $f \equiv 5(n \pmod{7}) \pmod{7}$, the same construction realizes ψ^n for every residue class; for $n \geq 67$ the special player need not participate in any F-triple, and for smaller n a

suitable number q of special triples is used. Thus a single parametric family (the W-family of Proposition 11 with the degree/f-triple data above) covers all n .

Corollary 16 (closed forms for the five remaining small values). For $n = 8, 10, 11, 12, 15$ explicit closed-form constructions are also found (via the same $\deg_G + r_i = D$ structure, with the data below; verified by formula (U)):

n	$n \bmod 7$	c	$ G $	$ F $	$D = \deg + r$ (non-special)
8	1	8	15	2	4
10	3	10	16	1	3
11	4	14	22	6	5
12	5	16	25	6	5
15	1	20	32	2	4

Each is a W-family game $v(G, F)$ with all swing counts $\propto (2, \dots, 2, 1)$ and $\text{Pairs}(F) \subseteq G$ (monotone), so $\text{Bz}(v) = \psi^n$.

Corollary 17 (full resolution — analytic). Combining Corollary 14 (analytic, via Lemma A and (6)) with Corollary 16, **every $n \geq 6$ admits an explicit closed-form simple game with $\text{Bz}(v) = \psi^n$** . The proof is fully analytic: for $n \equiv 0, 2, 3, 6 \pmod{7}$ the complement-of-edges construction of Proposition 8 realizes ψ^n for every admissible n (Lemma A: the two-value Erdős–Gallai conditions hold for all n in the residue class), and for $n \equiv 1, 4, 5 \pmod{7}$ the triangle construction of Proposition 13 with the closed form (6) realizes ψ^n for every $n \geq 22, 25, 19$ respectively; the finitely many remaining values $6 \leq n \leq 21$ are the explicit games of Corollary 16 ($n = 8, 10, 11, 12, 15$) and of Proposition 8 ($n = 6, 7, 9, 13, 14, 16, 17$), with $n = 18$ the $D = 4$ variant of Corollary 14. Kurz’s Conjecture 16 is therefore fully resolved with explicit constructions for all n , with no reliance on computer search for existence.

Verification of the core theorems (independent re-check)

The three central claims were re-verified independently of their original derivations:

(a) The W-family swing formula (U), Proposition 11. The identity $\beta_i(v(G, F)) = |G| + (n - 1) + |F| - 2 \cdot \deg_G(i) - 2 \cdot r_i + e_i$ was checked

against direct swing enumeration on 440 randomly generated monotone instances ($\text{Pairs}(F) \subseteq G$, $n = 4..14$): **0 mismatches**. The two special cases were checked separately — Theorem 1 ($F = \emptyset$, 199 random instances) and Theorem 2 ($G = \text{complement}(E)$, 29 instances) — also with **0 mismatches**.

(b) The closed-form constructions of Proposition 13 (analytic by Lemma A and (6)). As an independent confirmation of the analytic proof, the triangle-with-special-triples construction was generated and its swing counts verified via (U) for every $n \in [6, 50]$ with $n \equiv 1, 4, 5 \pmod{7}$ and $n \geq 18$, and for the complement-of-edges constructions for the other residue classes: **all verified** (the five values 8, 10, 11, 12, 15 are covered by the $\deg_G + r_i = D$ constructions of Corollary 16, also verified). The Erdős–Gallai conditions of Lemma A and the feasibility inequalities of (6) were additionally checked by direct computation for all n up to 10^6 — agreeing with the analytic conclusion.

(c) The necessary property of Proposition 10. Exhaustive enumeration for $n = 6$ (all 360 solutions) and infeasibility of the MILP with a “losing $(n - 1) - \text{set}$ ” constraint for $n = 7, 8$ confirm that all $(n - 1) - \text{sets}$ are winning in any game with $\text{Bz}(v) = \psi^n$.

All verification scripts are included in the reproducible code package.

Current coverage table

{ % do not increment counter

n	method	c
6	compl	4
7	compl	6
8	deg+r	8
9	compl	6
10	deg+r	10
11	deg+r	14
12	deg+r	16
13	compl	12
14	compl	16
15	deg+r	20
16	compl	12
17	compl	24

}

(18–75 see the tables in Proposition 12.) $n \equiv 0, 2, 3, 6 \pmod{7}$ are covered for infinitely many n by Proposition 8; $n \equiv 1, 4, 5 \pmod{7}$ are covered by the triangle construction of Proposition 13 and by Corollary 16 for the smallest values. **Every $n \geq 6$ has a closed-form construction.** Full explicit (G, F) data for $6 \leq n \leq 80$ are provided in machine-readable form in `appendix-data-n80.json`; entries for $41 \leq n \leq 80$ are produced by the closed-form constructions of Proposition 8 ($n \equiv 0, 2, 3, 6 \pmod{7}$) and Proposition 13 ($n \equiv 1, 4, 5 \pmod{7}$), and every entry ($6 \leq n \leq 80$) is re-verified by direct enumeration of the swing counts — equal to ψ^n , i.e. $(2c, \dots, 2c, c)$ — and of the monotonicity condition $\text{Pairs}(F) \subseteq G$.

Open problems

1. Can Proposition 10 (all $(n-1)$ – *sets* winning) be proved rigorously?
2. Is c necessarily even?
3. What is the minimal c for each n ? (Certified by the CP-SAT model of Section 7: $n=6$: 4, $n=7$: 6, $n=8$: 8, $n=9$: 6, $n=10$: 10.) For $n = 8$ the previously open “7 or 8” gap is closed: the infeasibility of every smaller even scale $c \in \{2, 4, 6\}$ is certified, and $c = 8$ is realized (Table 2); likewise $n = 9$ ($c \in \{2, 4\}$ infeasible, $c = 6$ realized) and $n = 10$ ($c \in \{2, 4, 6, 8\}$ infeasible, $c = 10$ realized). Note that c is even (Corollary 5), so the odd candidate $c = 7$ is already excluded; the certified values are conclusive because CP-SAT proves infeasibility exactly.
4. Is there a single uniform construction that realizes ψ^n for all n with no case distinction? The current resolution uses residue-specific parameters within one framework (Corollary 15).
5. (*Weighted case, Conjecture 15.*) Can the certified infeasibility of the exact target $(2c, \dots, 2c, c)$ for $6 \leq n \leq 10$ (Section 9) be turned into an analytic proof for all n ? In particular, the quantitative form asks for a universal $c > 0$ with $\|Bz(v) - \psi^n\|_1 \geq c/n$ for every weighted majority game v . The certified optimal distances $5/44, 0.085470, 0.066667$ ($n = 6, 7, 8$) give the empirical constant $c \leq 0.533$; an analytic argument with $c = \Theta(1)$ would settle the conjecture.

12. Application: an exact computational case study

We close with a concrete case study applying the two engines of Sections 10.1–10.3 to a real voting body: the weighted voting rules of the Council of the European Union (EU) and its predecessors. Every claim below is tagged with its epistemic status — **certified** (deterministic computation at a conclusive weight bound), **exact** (closed form / subset enumeration), or **contextual** (institutional background, not a theorem of this paper).

12.1 Actual Banzhaf power of historical EU Council bodies

The original European Economic Community (EEC) Council (Treaty of Rome, 1958) and its first enlargements were weighted majority games on $n = 6, 9, 10$ voters — exactly the certified range of Section 9. Their official qualified-majority weights and quotas (Felsenthal & Machover 1998, who tabulate the EU Council history; the accession treaties) and the exact Banzhaf power distributions are:

body	n	weights w	quota q	L_1 to ψ^n
EEC-6 (1958–1973)	6	$(4 \times 3, 2 \times 2, 1)$	12	$26/77 \approx 0.338$
EEC-9 (1973–1981)	9	$(10 \times 4, 5 \times 2, 3 \times 2, 2)$	41	$2136/5389 \approx 0.396$
EEC-10 (1981–1986)	10	$(10 \times 4, 5 \times 3, 3 \times 2, 2)$	45	$2528/6023 \approx 0.420$
EEC-12 (1986–1995) †	12	$(10 \times 4, 8, 5 \times 4, 3 \times 2, 2)$	54	$878/2323 \approx 0.378$
EU-15 (1995–2004) †	15	$(10 \times 4, 8, 5 \times 4, 4 \times 3, 3 \times 2, 2)$	62	$98402/255867 \approx 0.385$

Weights are in compact multiplicity form: e.g. 10×4 means four members of weight 10, a bare value means multiplicity one. The exact Banzhaf swing-count vectors β and the derived power distributions (percentages of total swings), in the same notation, are: EEC-6 $\beta = (10 \times 3, 6 \times 2, 0)$, power = $(23.8 \times 3, 14.3 \times 2, 0)\%$; EEC-9 $\beta = (53 \times 4, 29 \times 2, 21 \times 2, 5)$, power = $(16.7 \times 4, 9.2 \times 2, 6.6 \times 2, 1.6)\%$; EEC-10 $\beta = (100 \times 4, 52 \times 3, 26 \times 3)$, power = $(15.8 \times 4, 8.2 \times 3, 4.1 \times 3)\%$;

EEC-12 $\beta = (286 \times 4, 242, 148 \times 4, 102 \times 2, 40)$, *power* = $(12.9 \times 4, 10.9, 6.7 \times 4, 4.6 \times 2, 1.8)\%$; *EU-15* $\beta = (1968 \times 4, 1606, 1028 \times 4, 806 \times 3, 616 \times 2, 406)$, *power* = $(11.2 \times 4, 9.1, 5.8 \times 4, 4.6 \times 3, 3.5 \times 2, 2.3)\%$.

† *Beyond the certified range* ($n \geq 11$): exact infeasibility is not certified for these bodies (Tier-3 runs are heuristic, Section 9); the power distributions and L_1 values are exact computations reported as scaling context only, carrying no certificate.

Here L_1 is the distance to ψ^n minimized over which member occupies the half-power position of the egalitarian-with-one-exception target. Three observations:

- **EEC-6 has a null player.** Luxembourg (weight 1, quota 12 of total 17) has zero swings: no coalition of the remaining five members sums to weight 11, so Luxembourg’s vote is never decisive. Its power is 0, while the three large members (Germany, France, Italy) jointly hold 71.4% of all power. This is a known pathology of the original weighted Council, reproduced here exactly.
- **Real power is far from egalitarian.** The three bodies sit at L_1 – *distance* 0.338–0.420 from ψ^n , and the power spread (largest/smallest positive power) ranges from $3.85\times$ (EEC-10) to $10.6\times$ (EEC-9).
- **All three lie in the certified range** ($n = 6, 9, 10$), so the obstruction of Section 9 applies to them directly (exact infeasibility; the distance values above are exact subset-enumeration computations, reproduced by `case_study_eu.py` in the companion package).

12.2 What the certified machinery tells a designer

For each body with $n \leq 10$, Section 9 certifies — deterministically, at the conclusive weight bound $w_i \leq 2^{n-1}$ — that **no weighted majority game on n voters realizes ψ^n exactly**. The decision-support consequences for an institutional designer are:

1. **Certified infeasibility of the target.** No re-weighting of these councils — indeed no weighted rule on ≤ 10 voters at all — can attain the egalitarian target ψ^n . A designer who insists on a weighted rule must accept a strictly positive distance to ψ^n .
2. **Certified best achievable ($n \leq 8$).** Section 10.3 computes the exact minimal L_1 – *distance* over all weighted games: $5/44 \approx 0.114$ for $n = 6$ (attained at weights $(3,1,3,3,3,1)$, quota 11, and verified against exhaustive

enumeration of all 1,111 weighted games), 0.0855 for $n = 7$, and 0.0667 for $n = 8$, each certified as a global optimum over the full total-swing range. The actual EEC-6 rule sits at $L_1 = 26/77 \approx 0.338$ — so even within the weighted class, a redesign could improve the distance to ψ^6 by a factor of about three.

3. **Boundary of certification.** For $n \geq 11$ the exact infeasibility is not certified (Tier 3 runs are heuristic; Section 9). The later EEC-12 ($n = 12$) and EU-15 ($n = 15$) bodies lie beyond the certified range; their exact power distributions in Section 12.1 are reported as scaling context only, and their distance values carry no certificate.

12.3 The design decision and institutional context

The obstruction of Section 9 is a structural statement about the weighted class: egalitarian power is unattainable for $n \leq 10$, and the real EU bodies are far from it. This is consistent with the actual trajectory of the institution — **contextual** — the Council moved away from a single weighted quota in the Lisbon Treaty, adopting a double-majority rule (at least 55% of member states representing at least 65% of the population) that cannot be represented as one weighted majority game. The certified obstruction supplies the formal decision-support complement to that historical choice: it quantifies how far weighted rules can be from an egalitarian target, certifies that they cannot reach it for small bodies, and — via Section 10.3 — returns the exact best weighted approximation when the designer insists on staying in the weighted class.

12.4 A prescriptive design-analysis procedure

For a designer facing a target ψ on an n -voter body, the two engines of Section 10 compose into a decision-support procedure:

- if $n \leq 10$ and $\psi = \psi^n$, the Section 9 obstruction certifies the decision (infeasible in the weighted class; feasible by explicit construction in the general class, Section 5);
- if $n \leq 8$ and $\psi = \psi^n$, the Section 10.3 optimizer returns the certified best weighted approximation and its distance;
- otherwise the same solvers run under a time budget, and the outcome is labelled heuristic (Tier 3), not a certificate.

All numbers in this section are exact computations reproduced by `case_study_`

eu.py in the companion code package.

13. Conclusion

We complement the construction half with a negative result in the *weighted* universe (Section 9): the egalitarian target ψ^n cannot be reproduced by any weighted majority game. The obstruction is anchored in a classical swap/block identity (Proposition 4) for swing-count differences in weighted majority games, which forces the distinguished player to carry minimal weight and concentrates every other player’s deviation into a rigid subset-sum window; the sharp mirror form (M') of those windows — a new rigidity consequence of the classical swap/block identity — is our central structural contribution. Deterministic CP-SAT, at the conclusive weight bound $w_i \leq 2^{n-1}$, certifies infeasibility for $n = 6, 7, 8, 9$ and 10 ($n = 6$ consistent with the exhaustive enumeration of all 1,111 weighted games, whose optimal L_1 – distance to ψ^6 is $5/44$). We stress the distinction between proof and certification: the infeasibility of the exact target is a certified computation for $6 \leq n \leq 10$, not an analytic proof for all n , and likewise the quantitative form of Conjecture 15 — the universal bound $\|Bz(v) - \psi^n\|_1 \geq c/n$ — is certified for $n = 6, 7, 8$ (optimal L_1 distances $5/44, 0.085470, 0.066667$, giving $c \leq 0.533$), while the $\Theta(1/n)$ asymptotics for all n remains a conjecture supported by these certified values. The two-tier generalization $\psi^n(3, 1)$ is shown to be realizable by a *uniform* construction for all $n \geq 6$, while $\psi^n(k, 1)$ is obstructed for $k \geq 4$. This gives a complete picture of the two Kurz conjectures: Conjecture 16 holds by explicit construction, and Conjecture 15 holds *computationally* for $n \leq 10$, with the structural mechanism (M') explaining why — an analytic proof for all n remains open.

As an operations-research contribution, the paper offers a reusable, two-sided methodology rather than a one-off verification: a construction engine (Section 10.1) that is closed-form for every $n \geq 6$ and computationally verified for every $6 \leq n \leq 80$, and an exact verification engine (Section 10.2) that certifies non-existence at a conclusive weight bound, with the structural property (M') as its pruning engine. Both halves are fully reproducible from the companion code package, and both are explicit about their limits: the constructions cover every $n \geq 6$, while the weighted obstruction is certified for $n \leq 10$ and the quantitative constant $c \leq 0.533$ is certified for $n \leq 8$ only. The Section 12 case study grounds this methodology in a concrete institution — the weighted Council of the European Union — turning the abstract inverse-optimization

problem into prescriptive decision support for the design of voting rules.

Appendix. Additional explorations and negative results

The material in this appendix is kept for reproducibility and completeness; none of it is needed for the main results. It documents numerical observations and constructions that were explored and either remain open or did not realize the target.

Observation 7 (numerical). In all known solutions the scaling factor c is even ($c \in \{4, 6, 8, 10, 12, 14, 16, 22, 24, \dots\}$). Whether this is forced is open. (In the weighted case Corollary 5 resolves it: c must be even in any weighted realization.)

Observation 9 (hidden structure of MILP solutions). The MILP solutions for $n = 8, 10, 11, 12, 15$ (Section 7) all belong to the W-family (all $(n - 1) - sets$ win). Their threshold parameters (G, F) share: the special player has low degree in G (1–3) and appears in most F -triples; the graph G is connected. These reflect the solver’s specific choices and show no complete symmetry/cyclic structure, indicating that these n are “essentially mixed” and indeed cannot be covered by the complement-of-edges family (Proposition 8) or any single uniform family — consistent with the arithmetic description of Proposition 8. (Superseded for $n \equiv 1, 4, 5 \pmod{7}$ — including $n = 8, 11, 12, 15$ above — by the parametric triangle family of Corollary 15, which covers those residue classes at every n by varying q ; $n = 10$ ($n \equiv 3 \pmod{7}$) remains outside that family.)

Unsuccessful constructions and negative results.

1. **“Core-set” construction:** $v(S)=1 \iff |S| \geq n-1$ or $(|S| \geq n-3 \text{ and } C \subseteq S)$ for a fixed core C . For $n = 6..10$ no $|C|$ yields $\alpha(2, \dots, 2, 1)$. Reason: the core makes swings too symmetric.
2. **Random sampling of the threshold (E,F) family:** for $n = 8$ (known solvable), 4×10^6 monotone random (E,F) samples gave zero hits — the density of threshold-family solutions is essentially zero, so random search is hopeless and MILP is necessary. Recorded as a search lower bound.
3. **$F =$ all triples containing the special player:** makes the special player a null player ($\beta_n = 0$), so it does not realize ψ^n .
4. **$E = K_{n-1}$ minus a regular graph:** makes $\beta_n \equiv 0$, so it does not realize

ψ^n .

5. **Complement-of-edges at $n = 18$:** $n \equiv 4 \pmod{7}$, no integer (d,c) by Proposition 8. Confirmed.
6. **Closed-form parametric family for $n \equiv 1, 4, 5 \pmod{7}$:** CP-SAT was used to extract the full (G, F) solutions for $n = 22, 25, 26, 43$. These solutions are highly ad-hoc: the special player’s E-degree ranges from low (small n) to almost $n - 2$ ($n = 43$: degree 42), $|F|$ ranges from 4 to 299, and the c values are not unique across solvers ($n = 43$: HiGHS finds $c = 56$, CP-SAT $c = 362$, since c may be any multiple). **No clean parametric family could be extracted from these solutions**; this explains why these residue classes resist closed-form treatment (negative result, recorded). (Superseded: for $n \equiv 1, 4, 5 \pmod{7}$, Proposition 13 and Corollary 15 now derive the exact closed-form family from the deterministic certificates, so this negative conclusion no longer holds; the solver-dependence of the extracted c values observed here ($n = 43$: $c = 56$ vs 362) is explained there by the fact that c may be any multiple of the minimal feasible value.)

Declarations

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability. All results reported in this paper are reproducible from the companion code package (CP-SAT/HiGHS models, instance tables, and recorded runtimes) that accompanies this submission and will be released alongside the paper.

CRedit authorship contribution statement. Yongjin Chen: Conceptualization, Methodology, Investigation, Writing — original draft. Shi Jin: Formal analysis, Software, Data curation, Writing — review & editing. Songze Zhu: Conceptualization, Supervision, Writing — review & editing, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writ-

ing process. During the preparation of this work the authors used Claude (Anthropic) as an AI writing assistant to assist with language editing, typesetting, and formatting of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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